

Genomic insights into evolution of extensive drug resistance in *Stenotrophomonas maltophilia* complex

Sanjeet Kumar^{a,1,2}, Kanika Bansal^{a,2}, Prashant P. Patil^{a,3}, Amandeep Kaur^a, Satinder Kaur^b, Vivek Jaswal^c, Vikas Gautam^{b,**}, Prabhu B. Patil^{a,*}

^a Bacterial Genomics and Evolution Laboratory, CSIR-Institute of Microbial Technology, Chandigarh, India

^b Department of Medical Microbiology, Postgraduate Institute of Medical Education and Research, Chandigarh, India

^c Department of Cardiovascular and Thoracic Surgery, Postgraduate Institute of Medical Education and Research, Chandigarh, India



ARTICLE INFO

Keywords:

Extensive drug resistance

Stenotrophomonas maltophilia

Integrans

Complete genome

sul gene cluster

ABSTRACT

We report first complete genomic investigation of extensive drug resistance (XDR) in a nosocomial *Stenotrophomonas maltophilia* complex strain that is resistant to mainstream drugs (trimethoprim/sulfamethoxazole and levofloxacin). Comprehensive genomic investigation revealed its exclusive fourteen dynamic regions and highly enriched resistome comprising of two sulfonamide resistance genes on two diverse super-integrans of chromosomal origin. In addition, both these integrans harbour array of antibiotic resistance and commonly used disinfectant's resistance genes linked to ISCR elements. Isolation of a novel XDR strain from Indian tertiary care unit belonging to novel ST with diverse array of resistance genes on ISCR linked super-integrans indicates extent and nature of selection pressure in hospitals. Since, repetitive elements have major role in their spread and due to limitations of draft genomes, there is an urgent need to employ complete genome-based investigation for tracking the emergence of XDR at global level and designing strategies of antimicrobial stewardship and disinfection.

Importance: Hospital settings in India have one of the highest usages of antimicrobials and a heavy patient load. We hereby report a novel clinical isolate of *S. maltophilia* complex with two super-integrans that harbour array of antimicrobial resistance genes along with biocide and heavy metal resistance genes. Further, the presence of ISCR type of transposable elements on both the integrans indicates their propensity to transfer resistome while their chromosomal origin suggests possibilities for further genomic/phenotypic complexities according to selection pressure. Such complex mobile cassettes in a novel strain is a potential threat to global health care. Hence, to understand the evolution of opportunistic nosocomial pathogen, there is an urgent need to employ cost-effective long read technologies to keep vigilance on novel and XDR pathogens in populous countries. There is also need for surveillance of the usage of disinfectants and other antimicrobials for environmental hygiene and linked/rapid co-evolution of XDR in nosocomial pathogens.

Repositories: Complete genome sequence of *Stenotrophomonas maltophilia* SM866: [CP031058](https://doi.org/10.1016/j.ygeno.2020.06.049).

1. Introduction

According to World Health Organization (WHO) list, *Stenotrophomonas maltophilia* is one of the leading opportunistic multi-drug-resistance human pathogens in hospitals worldwide (http://www.who.int/drugresistance/AMR_Importance/en/). *S. maltophilia* causes a variety of infections including respiratory tract infections, bloodstream infections, bone and joint infections, urinary tract infections, endocarditis and meningitis [1]. *S. maltophilia* possesses intrinsic resistance to multiple groups of widely used antibiotics such as

[who.int/drugresistance/AMR_Importance/en/](http://www.who.int/drugresistance/AMR_Importance/en/)). *S. maltophilia* causes a variety of infections including respiratory tract infections, bloodstream infections, bone and joint infections, urinary tract infections, endocarditis and meningitis [1]. *S. maltophilia* possesses intrinsic resistance to multiple groups of widely used antibiotics such as

Abbreviations: Smc, *Stenotrophomonas maltophilia* complex; XDR, extensive drug resistance; ISCR, IS91-like elements common regions; ST, sequence type; TMP/SXT, trimethoprim/sulfamethoxazole; DR, dynamic regions

* Correspondence to: P.B. Patil, CSIR-Institute of Microbial Technology, Chandigarh 160036, India.

** Correspondence to: V. Gautam, Department of Medical Microbiology, Post Graduate Institute of Medical Education & Research, Chandigarh 160012, India.

E-mail addresses: r.vg@yahoo.co.uk (V. Gautam), pbpatil@imtech.res.in (P.B. Patil).

¹ Present address: CSIR- Institute of Himalayan Bioresource Technology, Palampur, India.

² Equal contribution.

³ Present address: Department of Microbiology, School of Medicine, University of Washington, Seattle, WA, USA.

<https://doi.org/10.1016/j.ygeno.2020.06.049>

Received 28 January 2020; Received in revised form 28 May 2020; Accepted 28 June 2020

Available online 10 July 2020

0888-7543/ © 2020 Elsevier Inc. All rights reserved.

cephalosporins, carbapenems, aminoglycosides and macrolides [2]. Trimethoprim/sulfamethoxazole (TMP/SXT) and levofloxacin are considered as the mainstream therapy for such infections [3,4]. Elevated reports of resistance to both of these drugs is a growing concern worldwide [1]. Hence, their rapid identification and susceptibility testing are key towards the successful treatment. *S. maltophilia* has various related species of diverse ecological origins that include clinical and environmental isolates forming *S. maltophilia* complex (Smc). It comprises of *S. maltophilia*, *Pseudomonas hibiscicola*, *P. beteli*, *P. geniculata*, *S. africana* and *S. pavanii*. The whole genome based studies of type and reference strains [5] belonging to Smc has revealed extensive and ongoing diversity in clinical isolates [6,7].

S. maltophilia has been reported to acquire resistance through integrons, transposons and plasmids conferring multiple resistance genes [9]. There are various PCR-based or draft genome based studies for the identification of these resistant strains [10–12]. ISCRs (IS91-like elements common regions), considered as gene-capturing systems of 21st century are found in the range of extensively drug resistant (XDR) bacteria carrying antibiotic and antimicrobial resistance genes [13,14]. For instance, in *Shewanella xiamenensis* the ISCR was linked with antibiotic resistance genes along with resistance genes for biocides heavily used in hospital settings such as quaternary ammonium compounds (*qac*), heavy metals (mercury resistance genes) etc. [15]. Interestingly, ISCR in *S. maltophilia* strains is also associated with *sul* genes. These are majorly reported from plasmids or with very low confidence, predicted to be chromosomally encoded [9,16]. Complete genome based studies of XDR *S. maltophilia* are critical to understand genomic context of the resistance gene cassettes. Such information is essential to understand genome dynamics particularly in new strains carrying an array of antibiotic resistance genes linked to ISCR elements [17].

Developing countries like India have high selection pressure because of heavy antibiotic and biocide usage, and high patient density coupled with tropical climate settings. Recent reports revealed rapid emergence of XDR in India in sulfonamide resistant *S. maltophilia* and carbapenem resistance conferred by New Delhi metallo-beta-lactamase 1 (NDM-1) in *Enterobacteriaceae* strains [1,18]. Furthermore, studies have reported the emergence of biocides (absolute ethanol, povidone iodine, sodium hypochlorite, and QACs) resistance among multidrug resistance bacteria such as *Escherichia coli*, *Klebsiella pneumoniae*, *Pseudomonas aeruginosa* and *Staphylococcus aureus* and diverse biofilm forming bacteria of dairy niche [19,20]. Use of emerging cost-effective nanopore technology for such extensive drug resistance will allow rapid access and further investigation of complete genome information.

In the present study, we report the isolation and complete genome based investigation of a novel sequence type (ST-366) of an extensively drug resistant strain belonging to Smc from a tertiary care center in Northern India. Complete genome level study of the strain reveals that it harbours fourteen dynamic regions (DRs). The strain also harbours two diverse chromosomal integrons linked to an array of antibiotic resistance genes along with insertion sequence (IS) elements and antimicrobial resistance genes. Overall, the study indicated serious long term consequences of high selection pressure for rapid emergence and evolution towards complete antimicrobial resistance strains in Indian hospitals. At the same time, the study highlights the need to employ rapid and long read technologies to carry out complete genome based investigations to understand and manage extensive drug resistance at global level.

2. Result

2.1. Bacterial isolation, identification and antimicrobial susceptibility

The strain SM866 was isolated in 2013 from blood specimen of an intensive care unit patient admitted to a tertiary care hospital in Northern India. This strain is also referred as SMGVX-1. Patient was admitted in the Cardiovascular and Thoracic surgery ICU on March

Table 1

SM866 zone of inhibitions for antibiotics: TMP-SMX: trimethoprim- sulfamethoxazole (25 µg), CHL: chloramphenicol (30 µg), LEV: levofloxacin (5 µg) and CAZ: ceftazidime (30 µg).

Antibiotics	SM866 (zone diameter in mm)	Result interpretation		
		(≥) Sensitive (S) (in mm)	Intermediate (I) (in mm)	(≤) Resistant (R) (in mm)
TMP-SMX	0 (R)	16	11 to 15	10
CHL	0 (R)	18	13 to 17	12
LEV	11 (R)	17	14 to 16	13
CAZ	0 (R)	21	18 to 20	17

2013. This patient was a 52 years-old gentleman with primary diagnosis of coronary artery disease – triple vessel disease, mild to moderate mitral regurgitation, dilated left ventricle with left ventricular ejection fraction of 30–35% and normal sinus rhythm along with comorbidities of diabetes mellitus, hypertension, asthma and hypothyroidism. The patient presented with New York Heart Association (NYHA) class III symptoms and underwent coronary artery bypass grafting. On 8th April, blood culture grew *S. maltophilia*, his total leukocyte count (TLC) was 17,200 with predominance of neutrophils (90%) on 12th April and he expired after 8 days due to septic shock and multiorgan dysfunction. SM866 exhibited 99.5% partial 16S rRNA nucleotide sequence identity with the type strain *S. maltophilia* ATCC 13637 (T). The 16S rRNA gene phylogeny using all the type strains of the genus *Stenotrophomonas* clearly showed that our isolate was in Smc (Supplementary Fig. 1(A)). For the phylogeny, we have considered all the known genomospecies of the Smc i.e. *S. maltophilia*, *Pseudomonas hibiscicola*, *P. beteli*, *P. geniculata*, *S. africana*, *S. pavanii*, SM5815, SM3123, SM10507 and SM16975 [7,]. Antibiotic resistance profiling revealed that SM866 was resistant to all the antibiotics tested, which included trimethoprim/ sulfamethoxazole, levofloxacin, chloramphenicol and ceftazidime (Table 1).

2.2. Multi-locus sequence typing and population structure analysis

Multi-locus sequence typing revealed that out of seven house-keeping genes (*atpD*, *gapA*, *guaA*, *mutM*, *nuoD*, *ppsA* and *recA*), SM866 harbours three novel housekeeping alleles i.e. *guaA*, *mutM* and *nuoD* with newly assigned allele numbers: 272, 147 and 141 respectively. Hence, SM866 was assigned as a novel sequence type ST-366. Further, population structure analysis with all the reported STs of *S. maltophilia*, clearly depicts that SM866 is a singleton ST and it does not belong to the clonal complex of any of the already known STs (Supplementary Fig. 1(B)).

2.3. Genome sequencing and annotation

The complete circular genome of SM866 with one chromosome (and no plasmid) of 5.08 MB with 60× coverage and 66.03% GC content was obtained by assembly of the Oxford Nanopore long reads and polishing with Illumina short reads. SM866 genome is submitted to NCBI with accession number CP031058. NCBI Prokaryotic Genome Annotation Pipeline (NCBI-PGAP) annotation resulted in 4715 CDS, 73 tRNA and 4 ncRNAs. Further, CheckM estimated 99.76% completeness and 0.72% contamination in the assembled genome. Plasmid profile of the strain SM866 was confirmed by plasmid screening, which resulted in no plasmid detection in SM866 and *S. maltophilia* ATCC13637 (T) (Supplementary Fig. 2(A)).

2.4. Phylogenomic and taxonogenomic analysis

Taxonogenomic analysis of SM866 was carried out to clearly know its species status. ANI and dDDH values for SM866 were below the cut-off values for species delineation. Within Smc, values ranged from

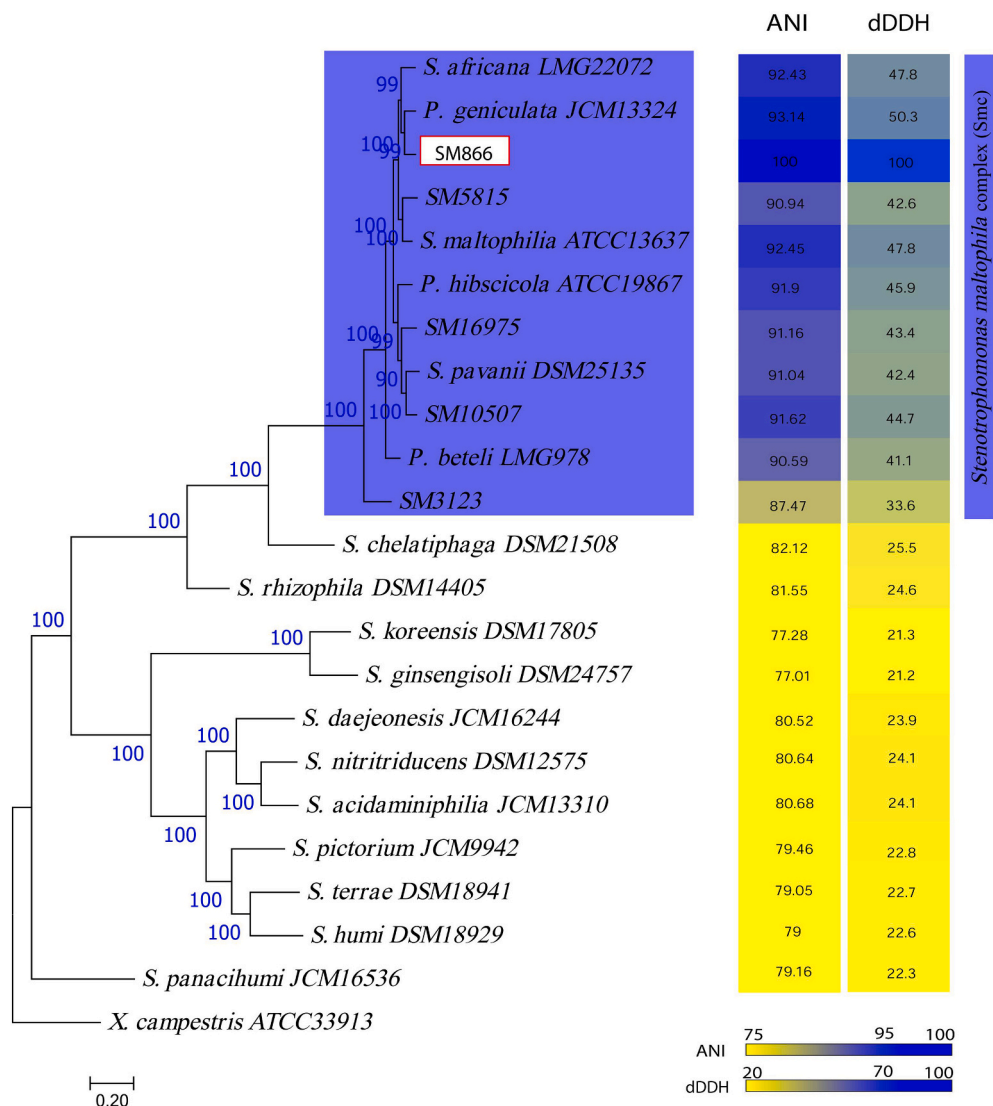


Fig. 1. PhyloPhlAn phylogeny and taxonogenomic analysis of representative strains of *Stenotrophomonas* genus along with SM866 (in red box) and Smc is highlighted in blue box. ANI and dDDH values for all the representative strains taking SM866 as reference are shown in the heat maps next to the tree generated on the scale as shown in the figure. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

87–93 % to 34–50% for (average nucleotide identity) ANI and (digital DNA–DNA hybridization) dDDH respectively (Fig. 1). Hence, this analysis suggested that SM866 is a potential novel species in Smc. Phylogenomic analysis of all the representative strains of *Stenotrophomonas* genus suggested that SM866 falls within the Smc complex (Fig. 1). Here, *Xanthomonas campestris* ATCC33913 (T) was used as an outgroup.

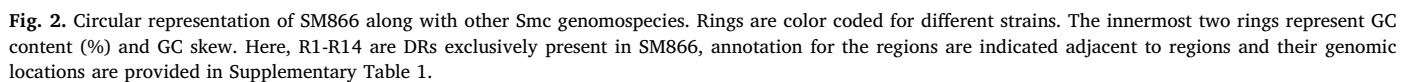
2.5. Understanding genome dynamics in evolution of extensive drug resistance

SM866 was found to be trimethoprim/sulfamethoxazole and levofloxacin resistant, unlike to previous known genomospecies of the Smc. Hence, we focused on the unique regions to SM866, which were absent from other Smc strains. Interestingly, we found 14 dynamic regions (R1–R14) exclusively in SM866. Functional classification of these dynamic regions along with their genomic locations is provided in Fig. 2 and Supplementary Table 1. Here, a range of antibiotic resistance genes (tetracycline, chloramphenicol, beta-lactamases, sulfonamide-resistance, aminoglycosides), efflux pumps (RND, ABC, cation transporter, Na⁺/H⁺ antiporter), heavy metal resistance (copper, mercury, arsenate), quaternary ammonium compound (QacE),

glycosyltransferases, methyltransferases, acetyltransferases, DEAD/DEAH box helicase, disulfide bond formation protein DsbA, DNA repair protein, transcriptional regulators (MerR, ArsR, XRE, CopG, Lrp/AsnC, LysR, TolC, TetR/AcrR, LuxR), toxin-antitoxin system HipA, lipoproteins, lysozyme, redox-sensitive transcriptional activator SoxR, integrases, sensor kinases, response regulators, IS elements, phage proteins, transposases, conjugal transfer proteins (TrbBCDEFGIJK, TraGJ) and proteases etc. were found in the regions unique to SM866.

2.6. Resistome analysis of Smc genomospecies

Resistome of Smc genomospecies is represented in Fig. 3. Among all the Smc genomospecies, only SM866 encodes *sul1* and *sul2* genes. Drug resistance genes for aminoglycosides, beta-lactams, chloramphenicol, fluoroquinolones; efflux pumps like RND, SMR, MFS, MATE and ABC transporters were examined to understand intrinsic antibiotic resistance SM866. Interestingly, in the vast resistome of SM866, *sul1* and *sul2* are exclusively in SM866 that further map to R11 and R9 dynamic regions respectively.



resistance markers and integron cassettes.

3. Discussion

4174

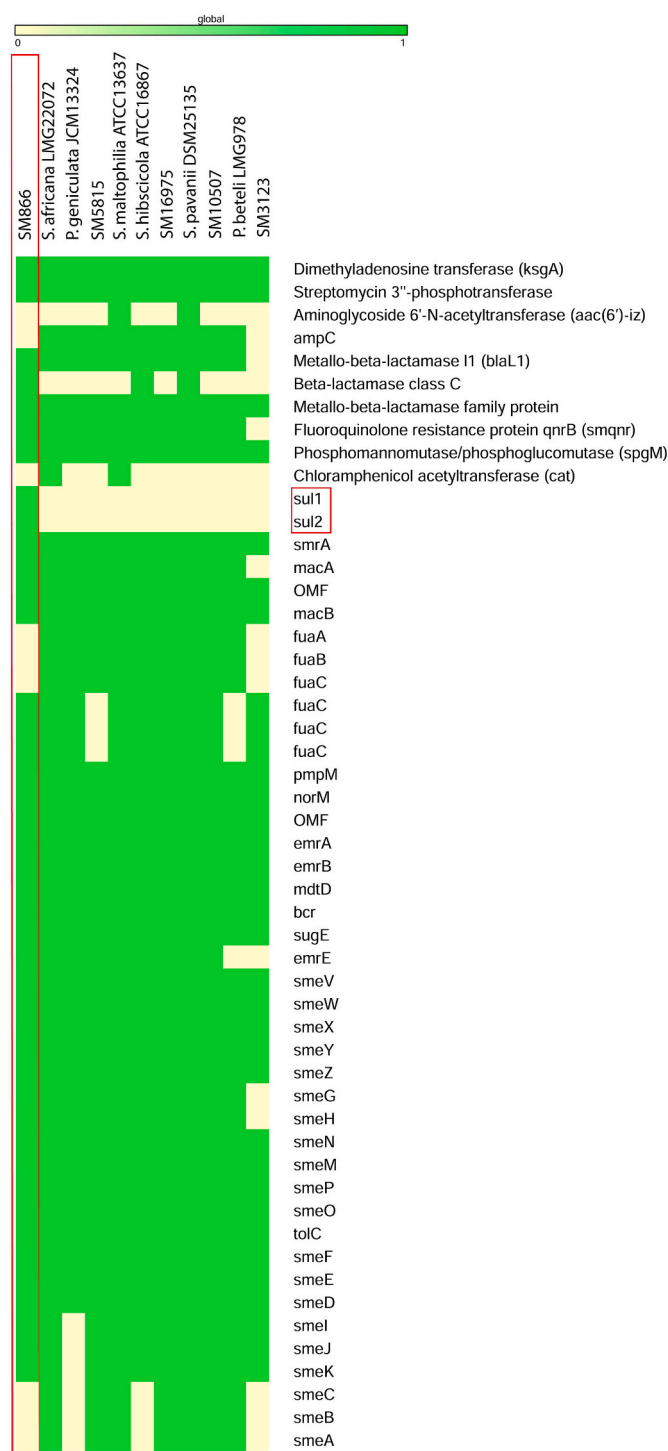


Fig. 3. Resistome heatmap of Smc distribution of antibiotic resistance and efflux pump genes across Smc. Here, presence and absence of gene is represented by green and yellow colors respectively. Here, SM866 is highlighted in red box. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

sul genes. At the same time, integrons carrying *sul* genes along with other commonly used antibiotics and biocide resistance in a highly evolved ISCR is alarming. To our knowledge, we are reporting the first complete genome sequence and investigation of an XDR strain from Smc [7,]. Hence, genome dynamics, considering the repetitive elements and mobilome, will be highly significant in understanding the evolution of extensive drug resistance in the tertiary care units of Indian hospitals.

Genome dynamics is the key for survival and evolution of the new-age superbugs. Heavy usage of disinfectants like quaternary ammonium compounds and heavy metals is a major activity in hospital settings [19]. Yet, hospital acquired infections with multi-drug resistant bacteria are persistent and prevalent. Hence, the effectiveness of cleanliness is always under question as the hospital settings are shown to be rapidly contaminated by deadly pathogens with multidrug resistant organisms like *Acinetobacter* spp., *Klebsiella pneumoniae* etc. [21,22]. These nosocomial pathogens have been shown to persist in the hospital environment from few days to several months [23]. In the present study, we hypothesize that it is this continuous disinfectants and antimicrobials usage leading to rapid emergence of an XDR strain by acquisition and intrinsic maintenance of resistance through multiple super-integrons linked to array of antibiotic resistance genes and ISCR elements. Hence, the heavy and continuous use of disinfectants might be leading to co-selection and linked resistance to array of antibiotics [24]. Such co-evolution and cross resistance of QAC-induced over-expressed efflux pumps towards antibiotics such as fluoroquinolones has already been reported in the literature [25,26]. Interestingly, biocide exposure of nosocomial pathogens has been reported to select for multidrug resistance [27]. In this context, our study goes on to reveal the extent, nature and potential for systematic and large scale antibiotic and antimicrobial resistance in a novel strain.

Complete genome dynamics of SM866 revealed exclusive DRs encoding for various efflux pumps, mobile elements, transcriptional regulators etc. Multidrug efflux transporters of RND family, MFS family, MATE and ABC transporters families etc. have already been reported in *Salmonella enterica*, *Mycobacterium tuberculosis* contributing towards the evolution of extensive drug resistance [28–30]. Presence of these determinants and also the fourteen DRs also point to tremendous selection pressure and rapid evolution of XDR Smc. The selection pressure can be attributed to high patient density in a developing country like India and heavy usage of antimicrobials contributing to extreme variation seen by two ISCR linked super-integrons at the chromosomal level that can also move across distant pathogens. Such a scenario further complicates and challenges the effective treatment of XDR strains among nosocomial pathogens.

Hence, in the present study, by investigation of the complete genome obtained using transformative long read technology, we are able to successfully reveal the cryptic but systematic and complex evolutionary radiation in the emergence of novel XDR strain within Smc. More such complete genome investigations of the superbugs at a global level are required for a better understanding of the clinical risks of reduced susceptibility towards biocides, which has got little attention than antimicrobial resistance. The present study emphasizes on awareness of targeted use of disinfectants restricting to the clinical benefit and not in an indiscriminate manner.

4. Methods

4.1. Bacterial isolation, identification and antimicrobial susceptibility testing

Bacterium was isolated from the blood sample of a hospitalized patient from Postgraduate Institute of Medical Education and Research (PGIMER), Chandigarh in 2013. Ethics approval and each patient's written consent was not required as it was a part of routine clinical testing. Bacterial cultures were grown in the nutrient broth at constant shaking of 200 rpm at 37 °C. DNA isolation was performed with ZR fungal/ bacterial DNA MiniPrep Kit (Zymo research) in accordance with manufacturer's instructions. In order to identify the bacterial species, partial 16S rRNA sequencing was performed using genomic DNA. Primers used for 16S rRNA amplification were 27F and 1492R. For Bacterial identification, EZ Biocloud [31] (<http://www.ezbiocloud.net/identify>) was used. Further, antimicrobial susceptibility testing was performed by the Kirby-Bauer disc diffusion method as per the Clinical

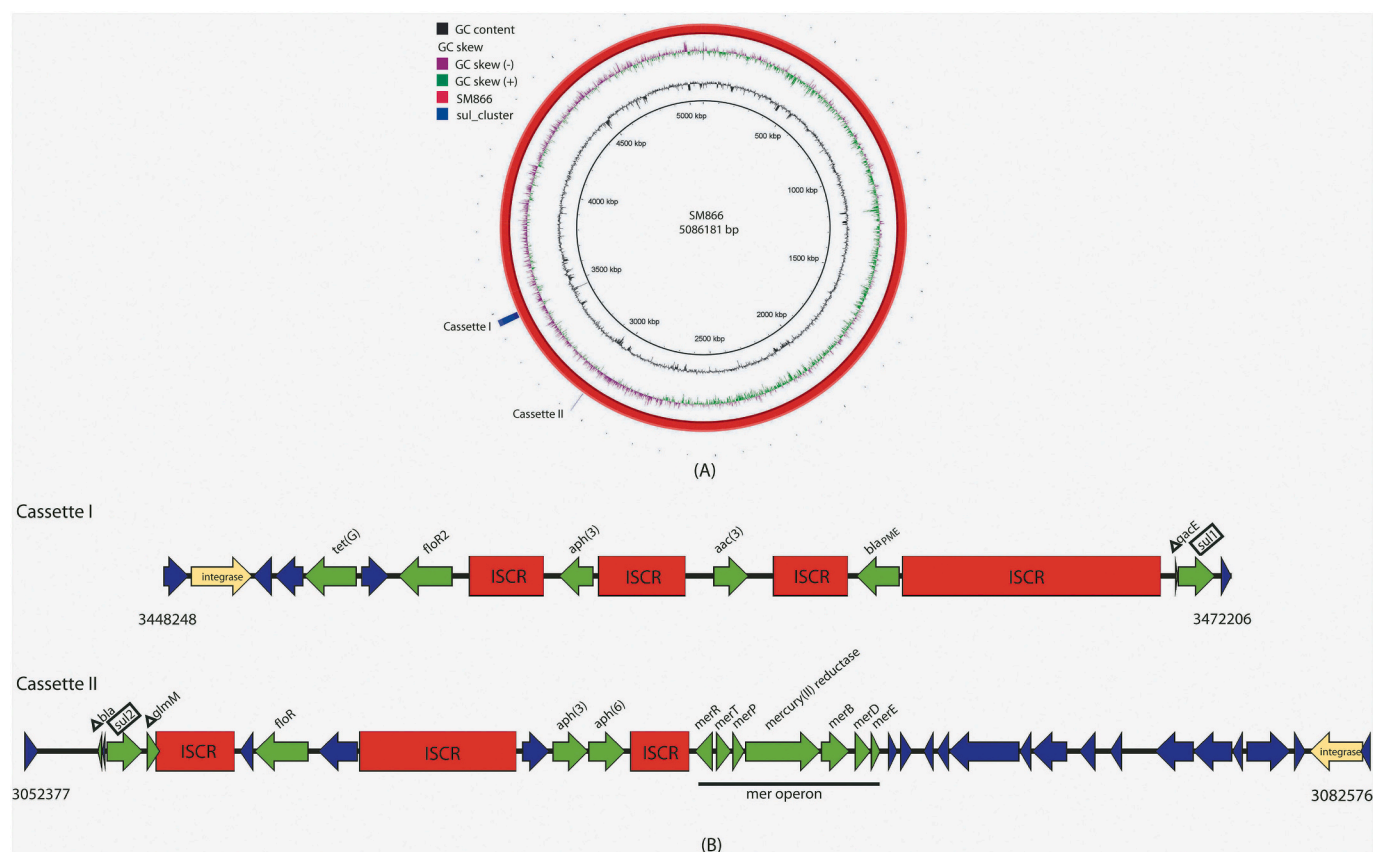


Fig. 4. (A) Circular representation of SM866 indicating chromosomal location of cassette I and cassette II. (B) Schematic representation of antimicrobial genes in cassettes I and II. ISCR elements are shown in red color and antibiotic resistance genes are in green color. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Laboratory Standards Institute (CLSI), 2020 guidelines against trimethoprim-sulfamethoxazole (25 µg), chloramphenicol (30 µg), levofloxacin (5 µg) and ceftazidime (30 µg) antibiotics.

4.2. Genome sequencing, assembly and annotation

4.2.1. Nanopore sequencing

Bacterial cultures were grown overnight in 15 ml of Luria-Bertani broth. Bacterial cells were harvested and high quality bacterial genomic DNA was isolated using Qiagen DNeasy Blood & Tissue Kit. DNA was purified using 0.45 × Agencourt AMPure XP beads (Beckman Coulter, High Wycombe, UK) and quantified using Nanodrop 1000 v3.8 (Thermo Fisher Scientific) and Qubit 2.0 (Thermo Fisher Scientific). The purified DNA sample was end-repaired (NEBnext ultra II end repair kit, New England Biolabs, MA, USA), cleaned up with 1 × AMPure XP beads (Beckman Coulter, USA). Native barcode ligation was performed with NEB blunt/TA ligase (New England Biolabs, MA, USA) and cleaned with 0.5 × AMPure XP beads. Qubit quantified and barcode ligated DNA samples were pooled at equi-molar concentration to attain 1 µg pooled sample. Adapter ligation (BAM adapters) was performed for 15 min using NEBNext Quick Ligation Module (New England Biolabs, MA, USA). Library mix was cleaned up using 0.4 × AMPure XP beads (Beckman Coulter, USA), and finally sequencing library was eluted in 15 µl of elution buffer.

4.2.2. Illumina sequencing

Genomic DNA was extracted from SM866 using ZR Fungal/Bacterial DNA MiniPrep Kit (Zymo Research, Irvine, CA, USA) and quantified using Qubit 2.0 Fluorometer (Life technologies). Short reads library was sequenced on Illumina MiSeq (Illumina, Inc., San Diego, CA, USA). Illumina sequencing library was prepared using Nextera XT sample

preparation kit (Illumina, Inc., San Diego, CA, USA) with dual indexing adapters and sequencing using 250 × 2 paired end sequencing kit. The adapters were trimmed during sequencing by Illumina MiSeq Control v.2.5.0.5 (Illumina, Inc., San Diego, CA, USA).

Genome assembly was first performed using the Unicycler v0.4.6 [32] with ONT long reads with bold mode. The complete genome obtained using ONT reads was then polished for multiple rounds of Pilon v1.22 [33] using Illumina raw reads of the sample SM866. The genome was assembled in a closed circular single chromosome. Annotation was performed using NCBI-PGAP [34] annotation pipeline. Assembly quality in terms of completeness and contamination was accessed using the CheckM v1.0.11 [35]. Genome coverage of the SM866 was calculated using BBMap v36.92 [36].

4.2.3. 16S rRNA gene phylogeny

Complete 16S rRNA gene sequence from the type strains of the genus *Stenotrophomonas* was fetched from their corresponding INSDC number (<http://www.insdc.org/>) from LPSN [37] given for their species definition (<http://www.bacterio.net/stenotrophomonas.html>). 16S rRNA sequence from SM866 was fetched from an assembled genome using a standalone version of RNAmmer v1.2 [38]. All the complete 16S rRNA gene sequences were aligned using ClustalW [39]. The phylogenetic tree was constructed based on neighbor-joining method with 1000 bootstrap. *Xanthomonas campestris* ATCC33913 [40] was used as an outgroup to genus *Stenotrophomonas*.

4.2.4. Multi-locus sequencing typing (MLST)

Sequence type analysis of the strain SM866 was done using center for genomic epidemiology MLST 2.0 (<https://cge.cbs.dtu.dk/services/MLST/>) and PubMLST (<https://pubmlst.org/bigdb/>).

4.3. Phylogenomic and taxonogenomics based identification and characterization

More robust phylogeny was constructed using > 400 conserved genes and using PhyloPhlAn [41]. *X. campestris* ATCC 33913 was used as outgroup. Further, taxonogenomic analysis of the strain was performed using orthoANI [42] and dDDH values [43].

4.4. Dynamic regions and resistome analysis

The unique DRs of SM866 were detected using BRIG v0.95 [44] which were further confirmed using standalone blastn package of BLAST+ v2.10.0 [45]. A well characterized nucleotide sequence of the antibiotic resistance gene(s) and efflux pumps were retrieved from the complete genome of *S. maltophilia* K279a [46]. A standalone BLAST was performed with these genes as a query with SM866 and the previously reported strains of Smc [7,]. A heat map showing the presence and absence of the gene cluster was generated using GENE-E v3.0.215 (<http://www.broadinstitute.org/cancer/software/GENE-E>). The schematic diagram of the cassettes I and II were generated using Easyfig v2.2.2 [47].

4.5. Plasmid profiling

The plasmid profiling was performed for isolate SM866 using FosmidMAX DNA Purification kit (Epicenter, Illumina). For this, the bacterial sample was grown in 10 ml of LB and incubated for 28 °C for 16 h. Bacterial cells were harvested by centrifugation at 8000 rpm for 5 min. Plasmid isolation was performed as per the manufacturer recommendations. 5 µl of the sample was loaded on 0.7% agarose gel to check the presence of plasmid (Supplementary Fig. 2). To reconfirm the absence of plasmid, Plasmid-Safe™ ATP-Dependent DNase was performed in accordance with manufacturer recommendation and *S. maltophilia* ATCC13637 (T) was used as a control.

5. Conclusion

Complete genome sequencing allowed us to establish conclusively that both the *sul* genes are of chromosomal origin in our XDR strain. Presence of two chromosomal ISCR elements linked super-integrations in a nosocomial strain indicated the importance of chromosomal route and extent of selection leading to the emergence of such a novel strain even in clinical settings. Gillings noted in his review that chromosomal integrations in contrast to transient plasmid borne integrations, can become a site to generate genomic and phenotypic complexity. In clinical setting, integrations are reported to be of plasmid origin and thought to be the result of human intervention [48]. In this context, the finding of a nosocomial pathogenic strain with two chromosomal integrations with complex genetic cassette suggests that clinical settings, like in environmental counterparts, have emerged as hotspots for integron evolution. Presence of antimicrobial genes in both the integrations of a novel sequence type suggests the nature of extreme selection happening in the hospital settings and further possibility of many more novel sequence types. In this context, finding of a novel ST with mutation in *mutM* gene is further worrisome. This is particularly relevant in tropical clinical settings like India, having high patient density and heavy use of antimicrobials. Hence, a coordinated global research effort by integrating emerging long read technologies is needed for surveillance of usage of antimicrobials and disinfectants, heavy metals etc. to halt linked/ rapid co-evolution of XDR strains among nosocomial pathogens.

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ygeno.2020.06.049>.

Ethical approval

Ethics approval and each patient's written consent was not required

as it was a part of routine clinical testing.

Author statement

V.G., S. Kaur and V.J. were involved in initial isolation and identification of the isolates. S. Kumar. and P.P.P were involved in complete genome sequencing. S. Kumar. and K.B. were involved in genome data analysis, interpretation and preparation of first draft of manuscript with inputs from all the authors. A.K. was involved in plasmid profiling. P.B.P and V.G. conceived the study and participated in its design with inputs from S. Kumar and K.B. All authors have read and approved the final manuscript.

Declaration of Competing Interest

The authors declare that there are no conflicts of interest.

Acknowledgements

All the work in the present study was supported by “GEAR- Genomic Evolutionary and Big-Data Analytic Strategies to address Antimicrobial Resistance” (MLP0016). We thank Dr. Manjunatha B. L. and other team members from Genotypic Technology (P) Limited, Bengaluru, India for assistance in obtaining sequencing data from Oxford Nanopore platform. We also thank Prof. Narottam Sharma, retired from Regional Institute of English, India for proofreading the article. The authors declare solemnly that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

References

- [1] P. Kaur, et al., Distribution of class 1 integrons, *sul1* and *sul2* genes among clinical isolates of *Stenotrophomonas maltophilia* from a tertiary care hospital in North India, *Microb. Drug Resist.* 21 (4) (2015) 380–385.
- [2] A. Nicodemo, J.G. Paez, Antimicrobial therapy for *Stenotrophomonas maltophilia* infections, *Eur. J. Clin. Microbiol. Infect. Dis.* 26 (4) (2007) 229–237.
- [3] S.A. Herrera-Heredia, et al., Risk factors and molecular mechanisms associated with trimethoprim-sulfamethoxazole resistance in *Stenotrophomonas maltophilia* in Mexico, *J. Med. Microbiol.* 66 (8) (2017) 1102–1109.
- [4] Y.L. Wang, et al., Monotherapy with fluoroquinolone or trimethoprim-sulfamethoxazole for treatment of *Stenotrophomonas maltophilia* infections, *Antimicrob. Agents Chemother.* 58 (1) (2014) 176–182.
- [5] P.P. Patil, et al., Genome Sequence of Type Strains of Genus *Stenotrophomonas*, 7 (2016), p. 309.
- [6] L.E. Ochoa-Sánchez, P. Vinuesa, Evolutionary genetic analysis uncovers multiple species with distinct habitat preferences and antibiotic resistance phenotypes in the *Stenotrophomonas maltophilia* complex, *Front. Microbiol.* 8 (2017) 1548.
- [7] P.P. Patil, et al., Taxonogenomics reveal multiple novel genomospecies associated with clinical isolates of *Stenotrophomonas maltophilia*, *Microb. Genom.* 4 (8) (2018).
- [8] R. Barbolla, et al., Class 1 integrons increase trimethoprim-sulfamethoxazole MICs against epidemiologically unrelated *Stenotrophomonas maltophilia* isolates, *Antimicrob. Agents Chemother.* 48 (2) (2004) 666–669.
- [9] L.-F. Hu, et al., *Stenotrophomonas maltophilia* resistance to trimethoprim/sulfamethoxazole mediated by acquisition of *sul* and *dfrA* genes in a plasmid-mediated class 1 integron, *Int. J. Antimicrob. Agents* 37 (3) (2011) 230–234.
- [10] A. Maravić, et al., Characterization of environmental CTX-M-15-producing *Stenotrophomonas maltophilia*, *Antimicrob. Agents Chemother.* 58 (10) (2014) 6333–6334.
- [11] S. Morsi, et al., Association of *sul* genes and class 1 integron with trimethoprim-sulfamethoxazole resistance in *Stenotrophomonas maltophilia* clinical isolates in Zagazig University, Egypt, *Afr. J. Clin. Exp. Microbiol.* 17 (3) (2016) 158–165.
- [12] P. Bennett, Plasmid encoded antibiotic resistance: acquisition and transfer of antibiotic resistance genes in bacteria, *Br. J. Pharmacol.* 153 (S1) (2008) S347–S357.
- [13] P. Boerlin, R.J. Reid-Smith, Antimicrobial resistance: its emergence and transmission, *Anim. Health Res. Rev.* 9 (2) (2008) 115–126.
- [14] K. Yousfi, et al., A novel plasmid, pSx1, harboring a new Tn 1696 derivative from extensively drug-resistant *Shewanella xiamenensis* encoding OXA-416, *Microb. Drug Resist.* 23 (4) (2017) 429–436.
- [15] M.A. Toleman, et al., Global emergence of trimethoprim/sulfamethoxazole resistance in *Stenotrophomonas maltophilia* mediated by acquisition of *sul* genes, *Emerg. Infect. Dis.* 13 (4) (2007) 559.
- [16] M.A. Toleman, et al., ISCR elements: novel gene-capturing systems of the 21st century? *Microbiol. Mol. Biol. Rev.* 70 (2) (2006) 296–316.
- [17] K.K. Kumarasamy, et al., Emergence of a new antibiotic resistance mechanism in

- India, Pakistan, and the UK: a molecular, biological, and epidemiological study, *Lancet Infect. Dis.* 10 (9) (2010) 597–602.
- [19] P. Gupta, et al., Emerging biocide resistance among multidrug-resistant bacteria: myth or reality? A pilot study, *J. Pharm. Bioallied Sci.* 10 (2) (2018) 96.
- [20] M. Makwana, et al., Biocides resistance profiles of biofilm forming bacteria of dairy niche and their control, *J. Homepage* 7 (02) (2018) 2018 <http://www.ijcma.com>.
- [21] A. Abdelaziz, et al., Exposure to sublethal concentrations of benzalkonium chloride induces antimicrobial resistance and cellular changes in *Klebsiellae pneumoniae* clinical isolates, *Microb. Drug Resist.* 5 (2019) 631–638.
- [22] D.R. Chung, et al., High prevalence of multidrug-resistant nonfermenters in hospital-acquired pneumonia in Asia, *Am. J. Respir. Crit. Care Med.* 184 (12) (2011) 1409–1417.
- [23] A. Kramer, et al., How long do nosocomial pathogens persist on inanimate surfaces? A systematic review, *BMC Infect. Dis.* 6 (1) (2006) 130.
- [24] S. Harbarth, et al., Is reduced susceptibility to disinfectants and antiseptics a risk in healthcare settings? A point/counterpoint review, *J. Hosp. Infect.* 87 (4) (2014) 194–202.
- [25] S. Buffet-Bataillon, et al., Efflux pump induction by quaternary ammonium compounds and fluoroquinolone resistance in bacteria, *Future Microbiol.* 11 (1) (2016) 81–92.
- [26] D.P. Gnanadhas, et al., Biocides–resistance, cross-resistance mechanisms and assessment, *Expert Opin. Investig. Drugs* 22 (2) (2013) 191–206.
- [27] M.A. Webber, et al., Parallel evolutionary pathways to antibiotic resistance selected by biocide exposure, *J. Antimicrob. Chemother.* 70 (8) (2015) 2241–2248.
- [28] M. Balganes, et al., Efflux pumps of *Mycobacterium tuberculosis* play a significant role in antituberculosis activity of potential drug candidates, *Antimicrob. Agents Chemother.* 56 (5) (2012) 2643–2651.
- [29] S. Sharma, et al., Piperine as an inhibitor of Rv1258c, a putative multidrug efflux pump of *Mycobacterium tuberculosis*, *J. Antimicrob. Chemother.* 65 (8) (2010) 1694–1701.
- [30] C. Willers, et al., Efflux as a mechanism of antimicrobial drug resistance in clinical relevant microorganisms: the role of efflux inhibitors, *Expert Opin. Ther. Targets* 21 (1) (2017) 23–36.
- [31] S.-H. Yoon, et al., Introducing EzBioCloud: a taxonomically united database of 16S rRNA and whole genome assemblies, *Int. J. Syst. Evol. Microbiol.* 5 (2016) 1613–1617.
- [32] R.R. Wick, et al., Unicycler: resolving bacterial genome assemblies from short and long sequencing reads, *PLoS Comput. Biol.* 13 (6) (2017) e1005595.
- [33] B.J. Walker, et al., Pilon: an integrated tool for comprehensive microbial variant detection and genome assembly improvement, *PLoS One* 9 (11) (2014) e112963.
- [34] T. Tatusova, et al., NCBI prokaryotic genome annotation pipeline, *Nucleic Acids Res.* 44 (14) (2016) 6614–6624.
- [35] D.H. Parks, et al., CheckM: assessing the quality of microbial genomes recovered from isolates, single cells, and metagenomes, *Genome Res.* 25 (7) (2015) 1043–1055.
- [36] B. Bushnell, BBMap: A Fast, Accurate, Splice-aware Aligner, (2014).
- [37] A.C. Parte, LPSN-list of prokaryotic names with standing in nomenclature (bacterio.net), 20 years on, *Int. J. Syst. Evol. Microbiol.* 68 (6) (2018) 1825–1829.
- [38] K. Lagesen, et al., RNAmmer: consistent and rapid annotation of ribosomal RNA genes, *Nucleic Acids Res.* 35 (9) (2007) 3100–3108.
- [39] J.D. Thompson, et al., Multiple sequence alignment using ClustalW and ClustalX, *Curr. Protoc. Bioinformatics* (2003) 1: 2.3. 1–2.3. 22.
- [40] A.R. da Silva, et al., Comparison of the genomes of two *Xanthomonas* pathogens with differing host specificities, *Nature* 417 (6887) (2002) 459.
- [41] N. Segata, et al., PhyloPhlAn is a new method for improved phylogenetic and taxonomic placement of microbes, *Nat. Commun.* 4 (2013) 2304.
- [42] I. Lee, et al., OrthoANI: an improved algorithm and software for calculating average nucleotide identity, *Int. J. Syst. Evol. Microbiol.* 66 (2) (2016) 1100–1103.
- [43] J.P. Meier-Kolthoff, et al., Genome sequence-based species delimitation with confidence intervals and improved distance functions, *BMC Bioinform.* 14 (1) (2013) 60.
- [44] N.-F. Alikhan, et al., BLAST ring image generator (BRIG): simple prokaryote genome comparisons, *BMC Genomics* 12 (1) (2011) 402.
- [45] C. Camacho, et al., BLAST+: architecture and applications, *BMC Bioinform.* 10 (1) (2009) 421.
- [46] B. Youenou, et al., Comparative genomics of environmental and clinical *Stenotrophomonas maltophilia* strains with different antibiotic resistance profiles, *Genome Biol. Evol.* 7 (9) (2015) 2484–2505.
- [47] M.J. Sullivan, et al., Easyfig: a genome comparison visualizer, *Bioinformatics* 27 (7) (2011) 1009–1010.
- [48] M.R. Gillings, Integrins: past, present, and future, *Microbiol. Mol. Biol. Rev.* 78 (2) (2014) 257–277.